

# **Composition Pair B1.15 + B2.26 — The Enabling Pair Where Action-Feedback Evolution Requires the Action Layer as Its Evidence Substrate, and the Action Layer Requires Action-Feedback Evolution to Give Its Accumulated Operational Records Governance Significance Beyond Compliance Archiving**

**Author:** Wenxin Li (Independent Researcher)

**ORCID:** 0009-0004-8065-3235

**Date:** May 12, 2026

**License:** Creative Commons Attribution 4.0 International (CC BY 4.0)

---

## **Attribution**

*This work derives from and formalizes the instinct/reasoning separation pattern introduced in The Instinct/Reasoning Separation Outside the Model (Li, April 2026), the second paper in the CKS theory series following Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026).*

*This note does not introduce new axioms. Its contribution is to formalize the composition pair formed by B1.15 (action-feedback evolution) and B2.26 (Action layer specification), specifying the enabling direction between them, their mutual dependency, the joint quality determination that follows, the co-presence requirement governing their correct deployment, and the failure mode that results when one is present while the other is absent or disconnected.*

---

## **Abstract**

The Coordination Knowledge Substrate (CKS) pattern, as extended in Paper 2 to cover AI Selves under human governance, commits to action-feedback evolution (B1.15) as one of three evolution mechanisms operating in productive tension. The mechanism depends architecturally on the Action layer (B2.26) as its sole evidence substrate: without Action layer records of actual cell operations, the proposing substrates that evaluate operational evidence have nothing to evaluate, and the two-stage human governance pathway governing action-feedback proposals has no content to process. This is the enabling direction — B2.26 enables B1.15. The significance direction runs the other way: without B1.15, Action layer records satisfy governance requirements for retraceability (A1.07) and provenance (A2.40) but serve no improvement purpose. They are compliance archives rather than intelligence inputs. B1.15 activates the Action layer's evolution significance, elevating its architectural role from passive record-keeping to active evidence evaluation. The pair is thus mutually dependent in a specific asymmetric structure: B2.26 provides the substrate B1.15 requires;

B1.15 gives B2.26 the significance that justifies its role beyond minimum governance compliance. The note formalizes this pair's structure, introduces the evidence quality dimension that makes Action layer record completeness architecturally important, and identifies Evidence Blindness (B3.28) as the primary failure mode when the enabling direction is severed.

## **1. Pair identification and relationship type**

**Pair:** B1.15 (action-feedback evolution — multi-shaped human governance over the operational-evidence-to-DNA-improvement cycle) + B2.26 (Action layer specification — the substrate record of cell task instances, carrying inputs received, outputs produced, instinct-layer processing decisions, and DNA consultation events per A2.40 provenance).

**Relationship type:** Enabling pair with significance reinforcement. The enabling direction runs from B2.26 to B1.15: the Action layer is the evidence substrate B1.15's machinery requires for its operation. The significance direction runs from B1.15 to B2.26: action-feedback evolution is what gives Action layer records a governance role beyond compliance archiving.

Neither direction subsumes the other. B2.26 describes what the Action layer contains and how it accumulates as cells operate; this commitment exists independent of whether action-feedback machinery is present. B1.15 describes the governed pathway from operational evidence to DNA improvement proposals authorized under two-stage human governance; this commitment requires an evidence substrate to function. The pair's composition requirement derives from both directions simultaneously: B1.15 cannot operate without B2.26's records; B2.26's records reach their full architectural significance only when B1.15's pathway evaluates them.

## **2. The enabling direction — B2.26 enables B1.15**

Action-feedback evolution per B1.15 evaluates accumulated operational evidence and, through governed proposing substrates, produces improvement proposals that advance through two-stage human governance before taking effect as directed selection events per B1.14. Every step in this pathway depends on Action layer records existing and carrying sufficient provenance per A2.40.

The proposing substrates that perform the evaluation operate on Action layer content: they examine records of inputs cells received, outputs cells produced, instinct-layer processing decisions cells executed, and DNA consultation events cells undertook. These records are what the proposing substrates identify as behavioral patterns, performance indicators, and systematic outputs that suggest DNA improvement opportunities. Without Action layer records, proposing substrates have no evidence base to evaluate. The Stage 1 governance review — which examines proposing substrate outputs for patterns worth advancing as proposals — would examine empty evidence. Stage 2 governance — which reviews specific proposals before approving them as substrate changes — would have no proposals to review. The entire action-feedback architecture, though present and correctly configured, would have nothing to operate on.

This is a clean enabling relationship: B2.26 provides the raw material that B1.15's machinery requires. The Action layer is not one possible source of evidence among several; it is the architectural location where operational records accumulate in the CKS pattern. There is no alternative evidence substrate for action-feedback to evaluate. The enabling direction is complete.

### **3. The significance direction — B1.15 gives B2.26 evolution significance**

Action layer records per B2.26 accumulate continuously as cells operate. Each task instance generates a record: inputs received, outputs produced, processing decisions taken, consultation events with the DNA layer. These records satisfy two minimum governance requirements regardless of whether B1.15 is present.

First, they satisfy A1.07 path retraceability. The six-field provenance vocabulary — writer attribution, timestamp, antecedent reference, rule reference, rationale where applicable, and relationship to contradicting content where applicable — applies to Action layer content as it applies to all substrate content. Operational records are traceable from their point of creation; the substrate carries the accountability vocabulary the governance architecture requires.

Second, they satisfy A2.40 provenance requirements for cell task instances. Each record's origin, processing context, and relationship to its antecedent substrate content is captured. The Action layer, in satisfying these requirements, plays a real and non-trivial governance role. Its records are not temporary; they constitute part of the substrate's authoritative state.

But both roles are passive. Records are created to satisfy governance requirements and stored for audit access. No evaluation of their content for improvement patterns is required by retraceability or provenance alone. The records accumulate; the deployment extracts compliance value; the improvement potential of the accumulated evidence is never realized.

B1.15 transforms this passive role into an active one. With action-feedback machinery present, the same records that satisfy A1.07 and A2.40 also serve as the evidence foundation for continuous operational improvement. Proposing substrates evaluate their content. Human governance reviews the resulting proposals. Approved proposals become directed selection events that change the DNA layer. The improved DNA produces different operational behavior, which generates different future Action layer records, and the cycle continues.

The architectural role of the Action layer is thus substantially different depending on whether B1.15 is present. Without B1.15: the Action layer is a governance-required record store, passive in its significance. With B1.15: the Action layer is an active intelligence input to the deployment's improvement, with the same records serving both compliance and improvement purposes simultaneously. B1.15 elevates B2.26's architectural significance from archive to evidence substrate.

### **4. The evidence quality dimension**

The enabling and significance directions together establish the pair's necessity: neither commitment delivers its full value without the other. The evidence quality dimension establishes a further architectural property: the quality of action-feedback proposals is jointly determined by the quality of Action layer records.

Action layer completeness per A5.08 provenance-completeness test is the relevant measure. A5.08 asks whether the provenance fields required to reconstruct an operational decision are present and accurate for the substrate content under evaluation. Applied to the Action layer, the test asks whether each task-instance record carries complete A2.40 provenance — whether the inputs, outputs, processing decisions, and consultation events are recorded with the attribution, timestamps, antecedent references, and rule references the accountability vocabulary requires.

Proposing substrates that evaluate Action layer evidence cannot compensate for incomplete records. If records are missing, proposing substrates evaluate an evidence base that does not accurately represent operational reality; the patterns they identify may not correspond to actual behavioral tendencies. If records lack provenance fields — if the rule under which a processing decision was made is not recorded, if the antecedent DNA content consulted is not referenced — the proposing substrate cannot distinguish between pattern classes that require different governance responses. The proposals that advance through Stage 1 and Stage 2 governance are only as informative as the evidence from which they were derived.

Incomplete Action layer records therefore produce degraded action-feedback proposals — proposals that are not merely lower in volume but potentially misleading in content. Governance may approve changes to the DNA layer that are not warranted by actual operational evidence, or may miss warranted changes whose supporting evidence was not captured. In either case, the improvement cycle operates on a distorted evidence base rather than on accurate operational intelligence.

This makes A5.08 provenance-completeness an architecturally significant test for the pair's health, not merely a standalone quality check. Maintaining Action layer completeness is a requirement for B1.15's proposal pathway to function with integrity. The pair's operational quality is jointly determined: complete provenance enables high-quality proposals; gaps propagate as degraded governance intelligence.

## 5. Co-presence requirement and failure states

Three states define the co-presence space for this pair.

**B2.26 without B1.15 — Action layer without action-feedback pathway.** Action layer records accumulate continuously as cells operate. The records satisfy A1.07 and A2.40. No proposing substrates are configured or governed. No evaluation of operational evidence is performed. Behavioral patterns in the evidence are never identified. Stage 1 and Stage 2 governance are never convened for action-feedback proposals. The improvement cycle the pair is designed to produce

never runs. Evidence Blindness (B3.28) precisely characterizes this state: the architecture has access to operational intelligence but exercises none of it. The Action layer grows; the deployment does not learn from what it accumulates. Operational context ages; the window in which evidence is informative about current behavior passes; the value of unread records decays.

**B1.15 without B2.26 — Action-feedback architecture without Action layer records.** Proposing substrates are configured and governed. Two-stage governance is in place. The pathway from evidence evaluation to DNA improvement is architecturally complete. But no Action layer records exist for proposing substrates to evaluate. Stage 1 governance reviews proposing substrate outputs generated from empty evidence. Proposals cannot be generated from nothing. The action-feedback machinery exists; it has no material to process. The cycle is architecturally present but operationally dormant.

**Full co-presence.** Cells operate, generating Action layer records per B2.26 with A2.40 provenance. Proposing substrates periodically evaluate Action layer content, identifying behavioral patterns, performance indicators, and systematic outputs suggesting DNA improvement opportunities. Stage 1 governance reviews proposing substrate outputs; promising patterns become proposals. Stage 2 governance reviews proposals; approved proposals become directed selection events per B1.14. The DNA layer improves. Improved DNA produces different operational behavior. Different future Action layer records accumulate. The cycle continues. The Action layer's evidence serves both its compliance function and its improvement function simultaneously.

## **6. Operational integration across the cycle**

The operational integration of the pair runs through a cycle with five identifiable phases.

**Accumulation.** Cells execute task instances. Each instance generates an Action layer record per B2.26 carrying the full A2.40 provenance vocabulary. Records accumulate in the substrate as cells operate, independent of whether evaluation machinery is running. This phase is entirely within B2.26's scope.

**Evaluation.** Proposing substrates periodically evaluate Action layer content. Evaluation identifies behavioral patterns in actual cell operations — what inputs produced which outputs under which orchestration rules, where processing decisions diverged from expected behavior, where DNA consultation events were frequent or sparse. The evaluation is governed: proposing substrates operate under orchestration rules that define evaluation scope, evidence selection criteria, and output format. This phase requires both B2.26 (the records to evaluate) and B1.15 (the proposing substrate machinery under governance).

**Stage 1 governance.** Human governance reviews proposing substrate evaluation outputs. Reviewers assess whether identified patterns represent genuine improvement opportunities, whether the evidence base is adequate, and whether a pattern warrants advancing to a formal proposal. Patterns that clear Stage 1 become proposals. This phase is within B1.15's governance architecture.

**Stage 2 governance.** Human governance reviews specific DNA improvement proposals. Reviewers assess the proposed change's content, scope, and expected effects. Approved proposals become directed selection events that modify the DNA layer. Rejected proposals are not enacted. This phase is within B1.15's governance architecture.

**Effect.** The improved DNA produces different orchestration behavior in subsequent cell executions. Future Action layer records reflect the changed behavior. The cycle returns to accumulation. The connection from approval to future evidence is what closes the loop architecturally.

Verification of the complete cycle's operation requires confirming that Action layer records are being evaluated by governed proposing substrates — not merely that records exist, and not merely that governance infrastructure is configured, but that the connection between the two is live and verified.

## **7. Detection and failure mode**

**Detection.** Three checks confirm the pair's correct operation. First, provenance-completeness of Action layer records per A5.08: are records complete enough to support quality evidence evaluation? Second, proposing substrate activity: are governed proposing substrates periodically evaluating Action layer content? Third, pathway connectivity: are evaluation outputs reaching Stage 1 governance, and are proposals from Stage 1 reaching Stage 2 governance? A deployment that passes A5.08 but shows no proposing substrate activity has the evidence but has severed the pathway. A deployment with active proposing substrates but degraded A5.08 provenance has the pathway but is feeding it incomplete evidence.

**Failure mode.** Evidence Blindness (B3.28) is the primary failure mode when B2.26 is present but B1.15's pathway is absent or disconnected. The Action layer accumulates operational evidence continuously. The evidence carries real intelligence about cell behavior, processing decisions, and DNA consultation patterns. But no evaluation machinery evaluates it. No proposals are generated. No governance reviews proposals. The DNA layer never receives the directed selection events that action-feedback evidence could support. The deployment's operational behavior does not improve through the feedback loop that its architecture should include. The evidence decays in governance value as operational context ages and the behavioral patterns the records document cease to represent current operation.

Evidence Blindness is not a temporary condition that self-corrects. Action layer records cannot trigger action-feedback evaluation on their own; the pathway from evidence to evaluation requires B1.15's proposing substrate machinery to be present and connected. A deployment in Evidence Blindness will remain in that state until B1.15's pathway is activated and connected to the accumulating Action layer content.

## **8. Operational test for the pair**

A deployment instantiates the B1.15 + B2.26 composition pair correctly if and only if all of the following are true:

1. Action layer records accumulate per B2.26 as cells execute task instances, carrying complete A2.40 provenance fields including inputs, outputs, instinct-layer processing decisions, and DNA consultation events.
2. Governed proposing substrates per B1.15 periodically evaluate Action layer content, producing outputs that identify behavioral patterns and improvement opportunities from the accumulated records.
3. Stage 1 governance reviews proposing substrate outputs and advances qualifying patterns as proposals.
4. Stage 2 governance reviews proposals and, where approved, enacts them as directed selection events per B1.14 that modify the DNA layer.
5. A5.08 provenance-completeness of Action layer records is maintained at a level that supports quality evidence evaluation — incomplete records that would produce degraded or misleading proposals are identified and addressed.
6. The pathway from Action layer accumulation through evaluation, Stage 1, Stage 2, and DNA modification is connected end-to-end — the existence of records and the existence of governance machinery are not sufficient; the connection between them must be live and verified.

A deployment in which (1) is satisfied but (2)–(4) are not instantiates B2.26 without B1.15 — the Evidence Blindness state. A deployment in which (2)–(4) are instantiated but (1) is absent or empty instantiates B1.15 without B2.26 — the dormant-pathway state. Only when all six conditions are met does the pair deliver its composed architectural property: operational evidence driving evidence-informed DNA improvement under two-stage human governance, closing the feedback loop from lived cell experience to governed structural change.

---

### Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

### How to cite this note

Li, W. (2026). *Composition Pair B1.15 + B2.26 — The Enabling Pair Where Action-Feedback Evolution Requires the Action Layer as Its Evidence Substrate, and the Action Layer Requires Action-Feedback Evolution to Give Its Accumulated Operational Records Governance Significance Beyond Compliance Archiving*. May 12, 2026. ORCID: 0009-0004-8065-3235.